

Motivating logics of indicative conditionals

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Outline

1. Logics of indicative conditionals
 - Twist representations
 - Example of application
 - Possible motivations
2. Truthmaker semantics
 - Residuated TMs
3. Motivating logics of indicative conditionals
 - Non-falsity-making
 - Exhaustive models
4. Future work

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For instance, we can preserve $\mathbf{1} \rightarrow \mathbf{0} = \mathbf{0}$ and $\mathbf{1} \rightarrow \mathbf{1} = \mathbf{1}$ but let $\mathbf{0} \rightarrow x := \top$. We also have to settle the remaining cases.

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$\rightarrow_{DF}$	0	T	1
0	T	T	T
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1	0	T	1

$\rightarrow_C$	0	T	1
0	T	T	T
T	0	T	1
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$\rightarrow_F$	0	T	1
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Consider the tables of Strong Kleene connectives plus Cooper's (Peirce's) quasi-conjunction and disjunction:

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$\vee_K$	0	T	1
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3. *Farrell's logic* F is the logic induced by $(\langle A_3; \neg, \wedge_K, \vee_K, \rightarrow_F \rangle, D)$,

Hence, these are logics of **non-falsity**: they are not truth-preserving but they do not introduce falsity.

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One normally wishes to prove a twist-representation result for the logic being studied.

This has been done for DF, C and F.

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Definition (DF twist-algebra)

Let $\mathbf{L} := (L; \wedge, \vee, 1)$ be an upper-bounded distributive lattice.

A *DF twist-algebra* over $\mathbf{L}$ consists in an algebra $\mathbf{L}^{\bowtie}$ with universe

$$\{(a_1, a_2) \in L^2 \mid a_1 \vee a_2 = 1\}$$

and operations

- i. $(x_1, x_2) \wedge (y_1, y_2) := (x_1 \wedge y_1, x_2 \vee y_2),$
- ii. $\neg(x_1, x_2) := (x_2, x_1),$
- iii. $(x_1, x_2) \rightarrow (y_1, y_2) := (x_2 \vee (x_1 \wedge y_1), x_2 \vee y_2).$

We can then define $x \vee y := \neg(\neg x \wedge \neg y).$

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We can then define $x \vee y := \neg(\neg x \wedge \neg y).$

This case is easy because we can equivalently add $\top := (1, 1)$ and define $x \rightarrow y := (\top \wedge \neg x) \vee (x \wedge y).$

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Definition (C twist-algebra)

Let $\mathbf{B} := (B; \wedge, \rightarrow)$ be a generalized Boolean algebra.

A *C twist-algebra* over $\mathbf{B}$ is the algebra $\mathbf{B}^{\boxtimes}$ having universe

$$\{(a_1, a_2) \in B^2 \mid a_1 \vee a_2 = 1\}$$

and operations:

- i. $(x_1, x_2) * (y_1, y_2) := (x_1 \wedge y_1, (x_1 \rightarrow y_2) \wedge (y_1 \rightarrow x_2))$,
- ii. $(x_1, x_2) \rightarrow (y_1, y_2) := (x_1 \rightarrow y_1, x_1 \rightarrow y_2)$,
- iii. $\neg(x_1, x_2) := (x_2, x_1)$.

The operation $+$ is obtained as the De Morgan dual of $*$, namely,

$$x + y := \neg(\neg x * \neg y).$$

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Definition (F twist-algebra)

Consider a generalized Boolean algebra $\mathbf{B}$.

An *F twist-algebra* over $\mathbf{B}$ as the algebra $\mathbf{B}^{\bowtie}$ with universe

$$\{(a_1, a_2) \in B \times B \mid a_1 \vee a_2 = 1\}$$

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- i. $(x_1, x_2) \wedge (y_1, y_2) := (x_1 \wedge y_1, x_2 \vee y_2),$
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- iii. $\top := (1, 1),$
- iv. $(x_1, x_2) \rightarrow (y_1, y_2) := (x_1 \rightarrow y_1, x_2 \vee y_2).$

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One thus obtains the four-valued logics DFg, Cg and Fg ($\perp$ is a *gap*).

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We have proven twist-representation results for these logics as well. **But what is the philosophical motivation of this approach?**

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... yay!

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$$(1a) \quad M, s \Vdash_e p \text{ iff } s \in V^+(p),$$

$$(1b) \quad M, s \nVdash_e p \text{ iff } s \in V^-(p);$$

$$(2a) \quad M, s \Vdash_e \neg\varphi \text{ iff } M, s \nVdash_e \varphi,$$

$$(2b) \quad M, s \nVdash_e \neg\varphi \text{ iff } M, s \Vdash_e \varphi;$$

$$(3a) \quad M, s \Vdash_e \varphi \wedge \psi \text{ iff } M, t_1 \Vdash_e \varphi \text{ and } M, t_2 \Vdash_e \psi \text{ where } s = t_1 \sqcup t_2;$$

$$(3b) \quad M, s \nVdash_e \varphi \wedge \psi \text{ iff } M, s \nVdash_e \varphi \text{ or } M, s \nVdash_e \psi.$$

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(1a) $M, s \Vdash_e p$ iff $s \in V^+(p)$,

(1b) $M, s \nVdash_e p$ iff $s \in V^-(p)$;

(2a) $M, s \Vdash_e \neg\varphi$ iff $M, s \nVdash_e \varphi$,

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(3b) $M, s \nVdash_e \varphi \wedge \psi$ iff $M, s \nVdash_e \varphi$ or $M, s \nVdash_e \psi$.

Define $\varphi \Vdash_e \psi$ as: for every TM M and every $s \in M$, $M, s \Vdash_e \varphi$ implies $M, s \Vdash_e \psi$.

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Similarly, we can define *inexact truth(falsity)making* as:

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It turns out that inexact truth(falsity)making for formulas is quite standard:

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$$|\varphi|_e^+ := \{s \in M \mid M, s \Vdash_e \varphi\}, \quad |\varphi|_e^- := \{s \in M \mid M, s \nVdash_e \varphi\};$$

$$|\varphi|_i^+ := \{s \in M \mid M, s \Vdash_i \varphi\}, \quad |\varphi|_i^- := \{s \in M \mid M, s \nVdash_i \varphi\}.$$

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We obtain:

$$|\neg\varphi|_e^+ = |\varphi|_e^-, \quad |\neg\varphi|_e^- = |\varphi|_e^+;$$

$$|\varphi \wedge \psi|_e^+ = |\varphi|_e^+ \sqcup |\psi|_e^+, \quad |\varphi \wedge \psi|_e^- = |\varphi|_e^- \cup |\psi|_e^-.$$

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We obtain:

$$\begin{aligned} |\neg\varphi|_e^+ &= |\varphi|_e^-, & |\neg\varphi|_e^- &= |\varphi|_e^+; \\ |\varphi \wedge \psi|_e^+ &= |\varphi|_e^+ \sqcup |\psi|_e^+, & |\varphi \wedge \psi|_e^- &= |\varphi|_e^- \cup |\psi|_e^-. \end{aligned}$$

And:

$$\begin{aligned} |\neg\varphi|_i^+ &= |\varphi|_i^-, & |\neg\varphi|_i^- &= |\varphi|_i^+; \\ |\varphi \wedge \psi|_i^+ &= |\varphi|_i^+ \cap |\psi|_i^+, & |\varphi \wedge \psi|_i^- &= |\varphi|_i^- \cup |\psi|_i^-. \end{aligned}$$

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Additionally, given $A, B \subseteq M$, we define:

$$s \in A \rightarrow B \text{ if and only if } \forall t \sqsupseteq s, t \in B.$$

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Suppose we identify a formula φ as the pair $(|\varphi|_i^+, |\varphi|_i^-)$.

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Heuristic: define inexact truthmaking conditions following the twist operations. Then, *exactify* them and find the corresponding models.

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An *exhaustive model* (EM) is a TM $(S, \sqsubseteq, V^+, V^-)$ in which:

(1a) $M, s \triangleright_e p$ iff $s \in V^+(p)$,

(1b) $M, s \triangleleft_e p$ iff $s \in V^-(p)$,

(2a) $M, s \triangleright_e \neg\varphi$ iff $M, s \triangleleft_e \varphi$,

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and where for every $s \in S$ and every formula φ it holds that $M, s \triangleright_e \varphi$ or $M, s \triangleleft_e \varphi$ (**exhaustivity condition**).

That is, an EM is a (negation only) TM in which every state is a non-falsity or a non-truth-maker of a given formula or both.

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We can define EMs for each target logic.

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A *DF exhaustive model (DFEM)* is a EM in which one defines:

(3a) $M, s \triangleright_e \varphi \wedge_K \psi$ iff $M, s_1 \triangleright_e \varphi$ and $M, s_2 \triangleright_e \psi$, where $s = s_1 \sqcup s_2$,

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A *C exhaustive model* (CEM) is a residuated EM in which one defines

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- (3b) $M, s \triangleleft_e \varphi \wedge \psi$ iff there is a map $\tau_1 : |\varphi|^+ \rightarrow |\psi|^+$ with $s_1 = \bigsqcup \{t \rightarrow \tau_1(t) \mid t \in |\varphi|^+\}$ and there is a map $\tau_2 : |\psi|^+ \rightarrow |\varphi|^+$ with $s_2 = \bigsqcup \{t \rightarrow \tau_2(t) \mid t \in |\psi|^+\}$, where $s = s_1 \sqcup s_2$,
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An *F exhaustive model (FEM)* is a residuated and intransigent EM in which one defines:

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Then, define $M, s \triangleright_e \neg \varphi$ iff $M, s^* \triangleright_e \varphi$.

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Can it be generalized to DF and F ?

References



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